

Natan Parmigiano

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SUMMARY

Principal Engineer and Software Architect with 8+ years of experience building distributed systems, cloud-native platforms, and real-time applications. Designed and operated 20+ production systems processing millions of transactions using Node.js, TypeScript, PostgreSQL, Kubernetes, AWS, and GCP. Strong track record driving architecture, platform reliability, and technical direction across growing engineering organizations.

EXPERIENCE

CTO | Paganini Tech | São Paulo | July 2021 - PRESENT

- Helped scale the organization from a 3-person operation to a 17-member engineering and delivery team.
- Architected and delivered more than 20 production systems spanning internal products, client platforms, and outsourcing initiatives.
- Established engineering standards and infrastructure patterns adopted across over 20 individual projects.
- Introduced observability, CI/CD, and platform engineering practices that became organization-wide standards.
- Designed and implemented distributed systems leveraging event-driven architectures, message queues, pub/sub patterns, CDC pipelines, and real-time data streaming architectures, processing millions of transactions and supporting large-scale real-time workloads.
- Led cloud infrastructure strategy across AWS and GCP using Docker, Kubernetes, Terraform, Pulumi, and CI/CD pipelines, maintaining over 15 production servers.
- Participated directly in hiring processes, performance management, roadmap planning, client meetings, and technical pre-sales initiatives.
- Continue to lead architecture reviews, incident response, infrastructure troubleshooting, and complex technical delivery initiatives.
- Primary stack: Node.js, TypeScript, Vue.js, PostgreSQL, Redis, Kubernetes, AWS, GCP, Docker, Pulumi, Grafana and AI Systems. Occasionally utilizing PHP, C#, Python, MySQL, DynamoDB, React, React Native.
- Directed organization-wide adoption of engineering best practices by conducting code reviews, implementing static analysis such as ESLint, successfully implementing practices across all 20+ projects over a 24-month period, in multiple tech stacks.

Senior Engineer | Paganini Tech | São Paulo | July 2020 - June 2021

- Architected and developed over 5 successfully deployed software products owning the entire stack from frontend to backend and SDLC and infrastructure.
- Led meetings with client stakeholders, acting as the contact point for defining and refining product requirements and validating delivery.
- Helped optimize and pinpoint performance issues in database queries and procedures measurably helping operational stability across different software solutions.
- Implemented and enforced SSL and deployment security practices into 4 legacy applications, handling the necessary adaptations and changes to bring a mature security posture to already deployed software.
- Successfully deployed, tracked operational performance and provided key touchstone training inside client organizations for more than 5 software solutions.
- Designed and implemented performance-critical components for high-volume real-time data processing systems, capable of processing thousands of simultaneous requests.
- Set up the initial cloud infrastructure and environment setup, personally configuring and deploying over 6 servers, that would become the foundation for the company's software operation.
- Weighed in on transformative stack changes that would set the new standards for future projects inside the organization, pushing the migration to market-leading stack choices and allowing the organization to measurably accelerate the software development process.
- Introduced and standardized PostgreSQL, Redis, Docker, Kubernetes, Pulumi, Grafana and other industry-leading technologies in an organization that utilized Oracle DB and Oracle APEX as the main stack, measurably reducing operational cost and complexity in new projects.
- Tackled client challenges in PHP, C#, Java, Python and mixed codebases, operating as the team's polyglot programmer and empowering an existing team to deliver value in unfamiliar grounds while keeping confidence high.

Co-founder & IT Director | Marketing Labs | São Paulo | August 2019 - July 2020

- Started an operation from scratch scaling to a 5 person team.
- Directly negotiated and landed contracts with over 5 clients spanning different sectors and building a strong network of contacts that would serve as a strong foundation throughout the years.
- Provided software development services to companies in the HR, Insurance, Real Estate and e-Commerce sectors, with services ranging from websites, e-commerce platforms, property management, sales portals and employee portals.
- Delivered 7 custom ERP systems, internal business applications, and end user facing applications.
- Created and owned the entire stack for solutions that reached large Brazilian organizations through strategic partnerships, 2 of which reached multinational corporations.
- Directly coordinated and handled technical execution, client relationships, and operational decision-making.
- Managed servers and infrastructure for all published projects.
- Implemented version control, SDLC, CI/CD and the entire development workflow for the organization.
- Primary Stack: PHP, MySQL, PostgreSQL, Javascript, frontend technologies (HTML, CSS).

Fullstack Developer | Make Acelerador de Vendas | São Paulo | March 2019 - September 2019

- Maintained and modernized legacy PHP applications.
- Architected and developed 2 custom ERP solutions in modern PHP frameworks.
- Developed pipelines for high-volume marketing data ingestion workflows and real-time reporting.
- Supported large-scale client platforms and mission-critical operational systems.
- Directly managed and implemented e-Commerce platforms for large enterprise clients.
- Direct ownership of the infrastructure and server configuration that supported the 3 applications developed during the contract.
- Managed the intranet solution for the company, including a 5 component internal shared storage, collaboration and task management suite developed in house based on an open source offering.

- Primary Stack: PHP, Mysql, Javascript, CSS, Vue.js.

Fullstack Developer | Freelancer | São Paulo | January 2017 - March 2019

- Delivered software development and technical consulting services for 4 clients across web applications, backend systems, and infrastructure environments.
- Developed and maintained applications using PHP, JavaScript, MySQL, and Flutter, supporting both new development initiatives and existing production systems.
- Diagnosed and resolved complex production issues involving application failures, deployment problems, server outages, and infrastructure misconfigurations across 3 clients.
- Performed Linux server administration, web server configuration, application deployment, and operational troubleshooting across diverse hosting environments for 4 live applications.
- Assisted businesses in modernizing legacy applications, recovering unstable systems, and restoring service continuity under time-sensitive conditions.
- Interacted directly with clients to understand technical requirements, define solutions, and deliver projects independently from conception through deployment.
- Built a reputation as a valuable resource for time-sensitive technical problem solving, often engaged to address high-impact issues under aggressive deadlines.

EDUCATION

B.Sc. Computer Science | ABC Federal University

São Bernardo do Campo, SP | JAN 2018 - FEB 2020

- Studied Computer Science at UFABC, one of Brazil's leading public universities for STEM education.
- Coursework focused on computer science fundamentals, software engineering, algorithms, data structures, and systems design, complementing parallel professional experience in software development and technology consulting.

Technical Diploma in Information Technology | ETEC Lauro Gomes

São Bernardo do Campo, SP | 2017

- Completed a highly competitive technical education program focused on software development, systems analysis, networking, databases, and information technology.
- One of the most competitive and renowned technical education institutions in the São Paulo region.

CERTIFICATIONS

Information Systems Auditing, Controls and Assurance

- Advanced coursework covering IT governance, risk management, auditing methodologies, controls, and assurance practices.

Data Engineering

Python for Data Science, AI & Development

Data Science

SKILLS

Software Architecture & Distributed Systems

- Distributed Systems Design
- Event-Driven Architecture
- Real-Time Systems
- Platform Engineering
- Microservices
- API Design
- Scalability & Reliability Engineering
- System Integration & Data Pipelines

Cloud & Infrastructure

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Kubernetes
- Docker
- Terraform
- Pulumi
- Infrastructure as Code (IaC)
- CI/CD Pipelines
- Observability & Monitoring

Backend, Frontend & Data Technologies

- Node.js
- TypeScript
- PostgreSQL
- Redis
- MySQL
- MongoDB
- PHP
- C#

- Vue.js
- Strong domain over web technologies (HTML, CSS, browser APIs)
- Pub/Sub Architectures
- Change Data Capture (CDC)
- Distributed Caching

Leadership & Professional Skills

- Technical Strategy
- Architectural Direction & Reviews
- Incident Response Posture
- Engineering Standards Proliferation & Evangelization
- Cross-Functional collaboration with non-technical teams
- Engineering Mentorship & Training
- Strong Technical Pre-Sales Experience
- Client Engagement
- Hiring & Team Development

Languages

- Portuguese — Native Fluency
 - English — C2 Proficient (Full Professional Fluency)
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